# Intensity, decibels and Doppler



**antocciphysics | TEACHER KEY | CPA + MATCHING SEI**

**Learning target:** Distinguish intensity change from frequency shift.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

**01. Warm-up: bring back an earlier idea** *Review: Unit 6, Day 8.*
Parallel oscillation; a longitudinal pressure wave.

**02. Worked example: explain the model**
Area proportional to $r^2$; total power conserved while intensity decreases.

**03. Read and interpret the model**
Higher on approach, lower on recession in basic Doppler model.

**04. Apply the physics**
+10 dB and +20 dB.

**05. Explain your reasoning**
No; intensity/amplitude and frequency/pitch are distinct.

**06. Review: connect an earlier idea** *Review: Unit 5, Day 11.*
24°C; ΔT is proportional to absorbed energy under these conditions.

**07. Exit check**
Higher in basic approach model.

**08. STRETCH: test the model**
1, 1/4, 1/9.

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**01. Warm-up: bring back an earlier idea** *Review: Unit 6, Day 8.*
150 cycles; T≈0.002 s.

**02. Worked example: explain the model**
Area proportional to $r^2$; total power conserved while intensity decreases.

**03. Read and interpret the model**
Higher on approach, lower on recession in basic Doppler model.

**04. Apply the physics**
6.02 dB; 80 dB intensity 100 times 60 dB.

**05. Explain your reasoning**
No; intensity/amplitude and frequency/pitch are distinct.

**06. Review: connect an earlier idea** *Review: Unit 5, Day 11.*
Absorbed 12000 J; source 240000 J.

**07. Exit check**
531.25 Hz.

**08. STRETCH: test the model**
Intensity ratio 1/9 gives -9.54 dB. Approach compresses arriving wavefront spacing/increases observed frequency; distinct from intensity change.